EXERCISE 7

BBX9Y2

I. Choose the best answer:

1. The factorial n is also written as

   (a) $n(n\text{-}1)!$ (b) $n(n+1)!$ (c) $(n\text{-}1)!$ (d) $(n+1)!$

2. The value of $10P_2$ is

   (a) 10 (b) 45 (c) 90 (d) 20

3. The number of different four letter words can be formed with the words 'DATE' is

   (a) 4 (b) 8 (c) 24 (d)48

4. The value of $50C_{50}$ is equal to

   (a) 50 (b) 25 (c)1 (d)0

5. The value of $20C_{18}$ is

   (a) 190 (b) 180 (c) 360 (d) 95

6. $\lim_{x \to 0} \frac{x^2-1}{x-1}$ is equal to

   (a) –1 (b) 0 (c) 1 (d) 2

7. Derivative of $\log x$ is equal to

   (a) 1 (b) $\frac{1}{x}$ (c) e^x (d) $\log x$

8. Derivative of x^9 is

   (a) x^8 (b) $9x^8$ (c) $8x^9$ (d) $8x^8$

9. The integral value of x^{11} is

   (a) $\frac{1}{12}x^{12}$ (b) x^{12} (c) $11x^{10}$ (d) $10x^{11}$

10. $\int e^x\,dx$ is equal to

    (a) e^{x^2} (b) e^x (c) e^{-x} (d) xe^x

11. $\int_0^1 x^3\,dx$ is

    (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) 3